

Text-to-SQL Dashboard

User Guide

Ask your database questions in plain English.
No SQL knowledge required.

How to use the application -- not a technical or setup reference.

What This App Does

This dashboard lets you ask questions about your database in plain English -- no SQL required. You type a question, the app writes the SQL query for you, shows it to you for review, and only runs it once you approve. Results appear as a table, with a chart drawn automatically when it makes sense.

Every query is read-only. The app can look at your data but can never change it.

A typical flow

1. Type a question, e.g. "Who are the top 5 customers by total sales?"
2. The app figures out which tables are relevant and writes a SQL query.
3. You review the SQL (and can edit it) before anything runs.
4. Click Confirm and Run to see the results as a table, plus a chart if applicable.

Starting the Application

From a terminal in the project folder, with the virtual environment activated:

```
streamlit run ui\app.py
```

The app opens in your browser, normally at:

```
http://localhost:8501
```

If the app won't start:

You'll see a red "Database Connection Required" screen instead of the normal app. This means the database couldn't be reached -- check with whoever manages your database connection settings before trying again.

The Sidebar: Database Connection

The left sidebar always shows the current connection at a glance:

- Type and Database -- which database engine and database name you're connected to.
- User -- your database login, partly masked for privacy (e.g. s***a).
- Test Connection button -- click any time to re-check the connection live and see a clear pass/fail message.
- Refresh Schema button -- re-scans the database's tables and updates what the app knows. Use this after your database's tables or columns have changed. It's quick if nothing actually changed.
- Discovered tables -- an expandable list of every table and column the app can see and use to answer your questions. Useful for checking whether the data you're asking about is actually available.

Asking a Question

Type your question into the chat box at the bottom of the main screen, in plain English, and press Enter. While the app is working, you'll see a status message describing what it's doing (retrieving schema, generating SQL, self-correcting if needed).

Tips for better questions

- Be specific about what you want: "total sales by year" works better than "show me sales."
- Name entities the way they appear in your data where you can (e.g. a category, region, or product name).
- If a question is broad or ambiguous, the app will make a reasonable assumption -- check the generated SQL to see what it decided, and re-ask more specifically if that wasn't what you meant.
- If you ask about something that isn't in the database at all, the app will say so rather than making up an answer.

Reviewing the Generated SQL

Before anything runs against your database, the app shows you:

- Retrieved schema context -- an expandable section listing which tables the app decided were relevant to your question.
- Generated SQL -- the actual query, shown in an editable text box. You can change it by hand if you want to adjust something.

Nothing runs automatically.

The SQL is only a proposal until you click Confirm and Run. Every click of that button -- including after you've hand-edited the SQL -- is checked again before it's allowed to execute.

Confirm and Run

Click Confirm and Run to execute the query. Results appear below as a table. If the shape of the data supports it (e.g. a category or date alongside a number), a chart is drawn automatically -- a line chart for trends over time, a bar chart for comparisons across categories. Not every result can be charted; a table-only result is normal and expected for some questions.

Large results are automatically capped at a maximum row count to keep the app responsive -- if you need a smaller, more specific slice of data, narrow your question (add a filter, a date range, or a "top N").

Understanding the Retry Timeline

Sometimes the first SQL attempt doesn't work -- a typo, a reference to the wrong column, and so on. When that happens, the app automatically tries again with the specific error in hand, up to a limited number of attempts, rather than giving up immediately.

Whenever more than one attempt happened, a Retry timeline section appears above the results. Expand it to see exactly what happened on each attempt -- useful if you want to understand why an answer took a moment longer, or why it didn't succeed at all.

What you might see in the timeline

- succeeded -- this attempt worked; its results are what's shown.
- syntax error / missing reference -- the SQL had a mistake or referenced the wrong table/column; the app corrected course and tried again.
- safety violation -- the app blocked itself from running something other than a read-only lookup. This never retries -- it's a hard stop, by design.
- timeout -- the query was taking too long and was stopped. This also doesn't retry automatically; see the next section for what to do.

What If It Fails?

If the app can't produce a working answer after its allotted attempts, you'll see a clear error message (not a blank screen or a crash), along with the last SQL it tried and a plain-language explanation of why it stopped. Common reasons:

- The question referenced data that genuinely isn't in the database.
- The question was too complex or ambiguous for the app to resolve confidently.
- The query was too expensive to run and timed out -- try narrowing the question (smaller date range, an added filter, fewer things compared at once).

Frequently Asked Questions

Can this ever change or delete my data?

No. Every query is restricted to read-only lookups before it's allowed to run.

Why does it sometimes take a few extra seconds?

That's usually the retry loop working -- see "Understanding the Retry Timeline" above. It's making sure you get a correct answer rather than a fast wrong one.

The sidebar says the database is disconnected. What do I do?

Click Test Connection to confirm and see the specific reason. If it's still failing, contact whoever manages the application's database configuration -- this is not something to fix from within the chat.

I asked the same question twice -- why was the second one instant?

The app remembers answers to identical questions within your current session, so repeating one doesn't call the AI model again.

Can I trust the chart?

The chart is a best-effort visual of the table above it -- always treat the table as the source of truth, and the chart as a convenience.